Capstone Design Project Report

Project Title : 소상공인 경영환경 개선을 위한 생성형 AI 기반 상업용 공간 이미지 생성 기능

Lecture Class	Capstone Design 2-01 (2025 Spring)			
Term	2025.03.04. ~ 2025.06.20			
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<Summary in English>

<table>
<tr><td>Project Goal and Objectives</td><td colspan="3">This project aims to develop HANCUT, a Diffusion-based AI system that allows users to easily generate interior design images using simple text descriptions and reference photos. The tool is designed for accessibility, making it usable even by non-experts. HANCUT integrates a Large Language Model (LLM), R-CNN, and a fine-tuned Vision-Language Model (VLM) to generate high-quality prompts tailored to interior design. These prompts are processed by a Diffusion model to produce realistic images. The web-based platform is publicly accessible, and both the source code and model weights will be open-sourced to encourage collaboration across industry and academia. By lowering the barrier to design, HANCUT supports both residential and commercial planning and contributes to the competitiveness of small business owners. The project ultimately aims to drive innovation in interior design through open-source sharing.</td></tr>
<tr><td>Project Contents</td><td colspan="3">This research consists of four sub tasks as belows.
【1st task】 Development of algorithms to generate Interior Prompts(LLM)
(1) Design an algorithm that generates interior-specific prompts based on user-provided spatial descriptions.
(2) Improve system prompt structuring and input processing using the GPT-4.1 nano API.
(3) Enhance prompt quality through preprocessing strategies and template-based approaches, and establish evaluation criteria.
【2nd task】 Development of R-CNN-based Object Detection to analysis Interrior-Related Informations
(1) Detect furniture and accessories in interior images using Faster R-CNN and return object types and confidence scores.
(2) Optimize dataset preprocessing and API response performance to improve real-time detection accuracy.
【3rd task】 Development of algorithm to generate an interior image as a prompt
(1) Extract style keywords from reference images using the SIGLIP model and incorporate them into the prompt generation process.
(2) Fine-tune the model using diverse style image datasets and release the trained weights as open source.
【3th task】 Development of a web-based platform
(1) Implement a web platform that integrates Sub-tasks 1 to 3, allowing users to generate interior images by inputting text and images.
(2) Use TypeScript and Material-UI for the frontend, and FastAPI for the backend to support real-time processing and image generation.</td></tr>
<tr><td>Anticipated Results and Impacts on Industry</td><td colspan="3">This study aims to develop AI technologies optimized for interior design image generation. By combining an LLM for prompt creation, R-CNN for image analysis, and a fine-tuned VLM, the system analyzes user-provided text and images to generate interior-specific prompts and detect furniture objects. The goal is to automate the design process and help small business owners communicate more efficiently with interior professionals. The source code and model weights will be open-sourced to support further AI development in the interior field. The research is expected to positively impact both the interior design industry and related markets, bringing practical changes through technology transfer to companies and individuals.</td></tr>
<tr><td rowspan="3">Keywords</td><td>LLM(Large Language Model)</td><td>VLM(Vision-Language Model)</td><td>R-CNN(Region Based Convolutional Neural Networks)</td></tr>
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<국문 요약>

<table>
<tr><td>과제의
목적과
목표</td><td colspan="3">본 과제는 누구나 손쉽게 인테리어 디자인을 시각화할 수 있도록, Diffusion 기반의 AI 생성 시스템(HANCUT)을 개발하는 것이다. 사용자는 간단한 인테리어 설명과 참조 이미지로 원하는 스타일과 객체가 반영된 인테리어 이미지를 생성할 수 있으며, 이는 비전문가도 손쉽게 설계할 수 있는 도구로 작용한다. 특히, 본 시스템은 사용자의 입력을 기반으로 LLM, R-CNN, 파인튜닝한 VLM 세 가지 AI 기술을 활용해, 인테리어 도메인에 적합한 고품질 프롬프트를 생성한다. 이 프롬프트는 Diffusion 기반 이미지 생성 모델에 입력되어 인테리어 이미지를 생성한다. 시스템은 웹 기반으로 구현되어 누구나 쉽게 이용 가능하며, 소스코드와 파인튜닝된 모델 가중치를 오픈소스로 공개해 산업계 및 학계와 기술을 공유하고자 한다. 이를 통해 상업 공간뿐 아니라 주거 공간까지, 설계 진입 장벽을 낮추고, 소상공인의 창업 및 매장 경쟁력 강화에 기여할 것으로 기대된다. 궁극적으로 HANCUT의 오픈소스화를 통해 인테리어 분야의 혁신을 촉진하는 것이 본 연구의 목표이다.</td></tr>
<tr><td>과제의
내용</td><td colspan="3">이 연구과제는 다음과 같이 네 개의 세부과제로 구성된다.
【1세부과제】 LLM 기반 텍스트 기반 인테리어 프롬프트 생성 알고리즘 개발
(1) 사용자의 공간 설명 텍스트를 바탕으로 인테리어 특화 프롬프트를 생성하는 알고리즘을 설계.
(2) GPT-4.1 nano API를 활용하여 시스템 프롬프트 구성과 입력 처리 방식 개선
(3) 전처리 전략과 템플릿 기반 접근으로 프롬프트 품질 향상 및 평가 기준 수립
【2세부과제】 R-CNN 기반 객체 탐지 및 인테리어 관련 정보 분석 알고리즘 개발
(1) Faster R-CNN을 통해 인테리어 이미지에서 가구 및 소품을 탐지하고, 객체 유형과 신뢰도를 반환
(2) 실시간 탐지 정확도 향상을 위해 데이터셋 전처리 및 API 응답 성능을 최적화
【3세부과제】 VLM 기반 이미지 기반 인테리어 프롬프트 생성 알고리즘 개발
(1) SIGLIP 모델로 참조 이미지의 스타일 키워드를 추출하고 프롬프트 생성에 반영
(2) 다양한 스타일의 이미지 데이터셋을 활용하여 모델을 파인튜닝하고, 가중치를 오픈소스로 공개
【4세부과제】 웹 기반 통합 플랫폼(HANCUT) 개발
(1) 1~3 세부과제를 통합해 사용자가 텍스트와 이미지를 입력하여 인테리어 이미지를 생성하는 웹 플랫폼 구현
(2) 프론트엔드는 TypeScript·Material-UI, 백엔드는 FastAPI로 구성하여, 실시간 분석 및 이미지 생성을 지원</td></tr>
<tr><td>예상되는
결과와
파급효과</td><td colspan="3">이 연구에서는 인테리어 디자인 이미지 생성에 최적화된 AI 프롬프트 생성 기술을 개발하고, 프롬프트 생성에 최적화된 LLM, 이미지 분석에 R-CNN, 파인튜닝한 VLM 모델을 기반으로 사용자가 제공한 텍스트 및 이미지를 분석하여, 인테리어 디자인 프롬프트 생성 및 가구 객체 탐지 기술을 개선할 것이다. 이를 통해 개인의 인테리어 디자인 프로세스를 자동화하고, 소상공인이 보다 효율적으로 인테리어 업체와 소통할 수 있도록 돕는 시스템을 구축할 예정이다. 또한, 소스코드와 파인튜닝된 모델 가중치를 오픈소스로 공개하고, 이를 통해 인테리어 디자인 분야의 AI 기술 발전에 기여할 것이다. 연구 결과는 인테리어 산업뿐만 아니라, 관련 시장의 확장에도 긍정적인 영향을 미칠 것이다. 또한, 본 연구에서 개발한 기술은 관련 기업 및 개인들에게 기술 이전을 통해, 국내 시장을 중심으로 인테리어 디자인 분야에서 실질적인 변화를 가져올 것으로 기대된다.</td></tr>
<tr><td rowspan="3">중요단어</td><td>LLM(Large Language Model)</td><td>VLM(Vision-Language Model)</td><td>R-CNN(Region Based Convolutional Neural Networks)</td></tr>
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1. Project Goals and Objectives (과제의 목표와 목적)

The detailed objective of the project has been established under the theme of "Demonstrating Generative AI-Based Commercial Space Image Generation and Editing Functions for Improving the Business Environment of Small Enterprises."

1.1 Project Goal (과제의 목적)

One effective way to improve the business environment for small enterprises is by enhancing the interior design of their commercial spaces. Traditionally, this process requires hiring an interior designer, receiving a proposal and estimate, and undergoing multiple reviews and revisions before construction. However, when the proposed design does not align with the business owner's expectations, additional changes may be needed, and they often result in higher costs and time delays.. This process can be particularly burdensome for small business owners with limited time and financial resources.

To alleviate this challenge, a practical solution is to provide designers with a clear visual reference or blueprint, allowing both parties to communicate more efficiently and reduce unnecessary revisions. In this context, generative AI can be used to automatically create commercial space images based on basic input, without requiring advanced design knowledge from the user.

The primary goals of this project are as follows:

A. To reduce the time required for generating and communicating interior design concepts.

B. To minimize unnecessary expenses associated with revisions and miscommunication.

C. To improve accessibility, enabling users without interior design expertise to visualize and present their ideas easily.

1.2 Project Objectives (과제의 목표)

한컷스튜디오(operated by My04) generates commercial space images using diffusion-based generative AI. Users upload space photos and design prompts to create visualizations that reflect their preferences, helping small business owners obtain design blueprints before engaging with professionals.

However, since API access to Onecut Studio's services was unavailable from the outset, the project has been redefined to develop an independent image generation system that operates entirely on user input. Our system accepts basic interior descriptions and reference images, then extracts relevant style and furniture elements using vision models such as a VLM (e.g., SIGLIP) and object detection models (e.g., Faster R-CNN). These elements, along with the user's input, are refined into a structured, domain-specific prompt using a Large Language Model (LLM). This prompt is then fed into a diffusion-based generative AI model to create

realistic interior design images, which are displayed through an interactive interface.

We refer to this complete end-to-end pipeline—from user input collection to image generation and visualization—as the “hancut” service.

The goal of hancut is to provide a practical tool that allows small business owners to visualize potential interior layouts and styles in advance, significantly reducing the time and cost of repeated consultations and revisions. Additionally, by improving usability and generation accuracy, the system enables even non-experts to easily produce relevant prompts and high-quality images. Ultimately, hancut aims to make interior design more accessible, intuitive, and efficient for the general public.

1.2.1. Objectives and Evaluation Criteria

To quantitatively assess whether the system reflects user preferences, we apply the same AI models to both the input reference image and the generated output. This enables an objective comparison of how well visual elements are retained throughout the image generation process.

Style Consistency Accuracy evaluates the alignment of stylistic elements between the reference and generated images. We use an image-text model to extract style tokens from both images, and calculate accuracy using the formula:

- Style Accuracy (%) = (Matched styles) / (Total styles in reference image) × 100.

For example, if 3 out of 4 extracted styles are present in the output, the result is 75%. The threshold for success is set at 50 percent, reflecting the fact that image style may be influenced by user prompts more than object content.

Object Consistency Accuracy measures whether major interior items such as furniture are preserved in the generated image. We detect objects in both images and apply the formula:

- Object Accuracy (%) = (Matched objects) / (Total objects in reference image) × 100.

If 2 out of 3 items match, the accuracy is 66 percent. A higher success threshold of 60 percent is required to ensure that essential elements are not lost.

Together, these metrics help verify that the generated images reflect both stylistic intent and key content. The style threshold is set slightly lower than the object threshold because user input can reduce visual style alignment with the original reference.

2. Background of the Project (과제의 필요성)

2.1 Motivations (필요성)

- Technical Necessity

Generating interior design images with AI requires well-structured prompts, but users without design expertise often struggle to express their preferences clearly. To address this, a Vision Model extracts key design elements—such as furniture and style—from reference images, simplifying prompt creation. A Large Language Model (LLM) then combines these elements with user input to form a coherent interior-specific prompt. This helps the diffusion model generate more accurate and stylistically consistent images, even from vague or fragmented descriptions.

- Economic Necessity

Small business owners often face conflicts and extra costs due to unsatisfactory design proposals caused by miscommunication. Using prompt engineering based on LLMs and vision models, contractors can receive clear, AI-generated design blueprints, reducing misunderstandings, time, and costs.

For DIY enthusiasts, this service lowers barriers to creating design plans without professional help, potentially stimulating the interior market by encouraging purchases of furniture and decor.

- Social Necessity

Home decoration as a hobby offers comfort and accomplishment. This service simplifies design blueprint creation, promoting the trend of home decoration and enhancing societal well-being by providing a creative and accessible outlet.

2.2 Significances (중요성)

- Importance from a Market Perspective

To grow the interior design market, contractors must deliver proposals aligned with customer preferences. Generative AI enables customers to create design blueprints that clearly communicate their needs, reducing disputes and improving satisfaction. This creates additional demand and expands the market by simplifying the design process for all users.

- Importance from a Technical Perspective

The quality of AI-generated images heavily depends on prompt quality. By developing and sharing diffusion-based generative AI prompts optimized for interior design, non-major

companies can enhance their service quality and competitiveness.

- Importance from a Social Perspective

Home decoration, often limited by design expertise, becomes more accessible with an interior prompt service. This technology allows anyone to create DIY blueprints, fostering a stronger connection to their living spaces and promoting emotional well-being.

3. Related Works (관련연구)

3.1 Related Techniques

3.1.1. GPT-4.1 nano

GPT-4.1 nano is a lightweight large language model optimized for fast, low-resource environments such as mobile or embedded systems. While efficient, its standalone use may lead to vague or inconsistent outputs when processing fragmented or unstructured prompts.

While GPT-4.1 nano alone may produce inconsistent results, our system integrates it with a structured prompt framework that organizes user descriptions, style tokens, and furniture tokens into a coherent interior prompt. This significantly improves semantic precision and ensures consistent formatting. The method also avoids the resource costs associated with larger LLMs, achieving both efficiency and output reliability. As a result, Hancut can deliver high-quality outputs with minimal computational burden, increasing accessibility for users without advanced hardware.

3.1.2 SIGLIP Fine-Tuning

SIGLIP is a vision-language model that improves upon CLIP by using sigmoid loss, framing image-text alignment as a binary classification task. This approach improves learning efficiency and reduces the risk of misalignment. Compared to CLIP, SIGLIP is more effective at filtering out vague or irrelevant styles, which is essential for precise style interpretation.

In the Hancut service, the model is fine-tuned on interior design images to enhance its performance in identifying interior-specific styles. Although SIGLIP demands more computational resources, the fine-tuned version used in Hancut delivers significantly better results than general-purpose models like CLIP in accurately recognizing interior styles. This trade-off prioritizes precision over speed, making it well suited for applications where accuracy is more important than processing time.

3.1.3. Faster R-CNN

Faster R-CNN is a widely used object detection model capable of classifying and localizing objects within images. However, it has limitations in precisely distinguishing overlapping items or objects in visually complex scenes, as it relies on rectangular bounding boxes. Despite this, its fast inference speed and compatibility with CPU environments make it well-suited for lightweight applications like the Hancut service.

While less precise in dense layouts, Faster R-CNN delivers detection results within seconds without GPU acceleration. Moreover, the availability of pre-trained models allows for easy integration and rapid deployment without additional training, reducing infrastructure burden and enhancing overall efficiency.

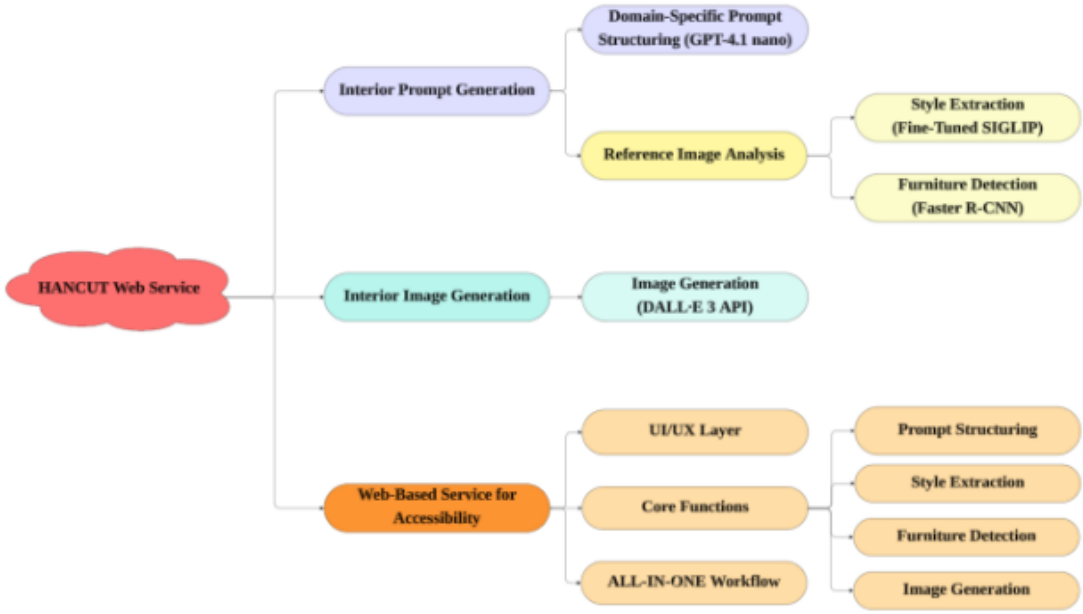
3.1.4. DALL·E 3 API

DALL·E 3 is a generative AI model that creates high-resolution images from text prompts, effectively capturing complex scenes and contextual elements. Although it may not reproduce ultra-fine styles with full accuracy, it excels at generating coherent images from multi-object descriptions.

For the CPU-based Hancut service, DALL·E 3's API-based architecture offers significant benefits: it eliminates the need to host or fine-tune a local model, reduces infrastructure and training costs, and minimizes server load by enabling on-demand usage. This makes it a scalable, cost-effective, and convenient alternative to self-hosted diffusion models.

4. Requirement Analysis (요구 분석)

4.1 Objective Tree (목적나무)



[Figure 1] Objective Tree of the HANCUT Web Service System

In the [Figure 1], the project is divided into three core components: interior prompt generation, image generation, and web-based service delivery. Prompt generation combines user text input (GPT-4.1 nano) and reference image analysis (SIGLIP and Faster R-CNN). Image generation is performed via the DALL·E 3 API using structured prompts. The web service component integrates interface design and core functions to support an accessible and unified user experience.

4.2 Functional Requirement Analysis (요구 분석)

4.2.1 Function-Related Requirement (기능 관련 요구사항)

The domain-specific interior prompt generation feature should produce structured prompts based on user-provided text, capturing key interior elements like style and layout. This will be done using GPT-4.1 nano, optimized for generating prompts suitable for image synthesis models.

The reference image style extraction feature will use a fine-tuned SIGLIP model to identify dominant interior styles (e.g., modern, Scandinavian) from uploaded images. The extracted style tokens will be integrated into the final prompt for consistent image generation.

The object detection feature will identify major furniture and layout elements in reference

images using a pretrained Faster R-CNN model. Detected object data (type, position, size) will be structured and shared with other modules for prompt refinement.

The image generation feature will use the DALL·E 3 API to create realistic interior images based on the generated prompts. The model will combine user input and extracted elements to produce visually coherent results.

The web-based platform will provide an intuitive interface where users can input descriptions or upload images, then view prompts, analysis results, and generated images in real time. The UI/UX must support easy access and smooth interaction for users of all experience levels.

4.2.2 Data-Related Requirement (자료 관련 요구사항)

The system handles user-provided data through a structured pipeline to generate customized interior design images. Inputs consist of a short natural language sentence describing the user's design preferences and a URL linking to a reference image in .jpeg, .png, or .webp format. Only URLs are accepted to ensure secure and lightweight processing.

Once received, the data is analyzed in real time. A fine-tuned SIGLIP model extracts the top three style tokens from the image, and a pretrained Faster R-CNN detects up to five furniture items with at least 70% confidence. These results are merged with the user's text and processed by GPT-4.1 nano to generate a domain-specific prompt. The final output includes the prompt and a DALL·E 3-generated image URL, both returned as strings.

Each request results in one prompt and one image, which are immediately displayed through the interface. No data is stored on the server after rendering. Limited metadata may be used for session-level diagnostics but is not retained or linked to personal user information, ensuring both privacy and system efficiency.

4.2.3 Interface-Related Requirement (인터페이스 관련 요구사항)

The system must provide an intuitive interface that enables users to submit text descriptions and image URLs as input. All inputs are sent to the backend via RESTful APIs and processed in real time to ensure responsiveness.

Using a fine-tuned SIGLIP model, the system extracts the top three interior style tokens from the reference image. A pretrained Faster R-CNN model detects up to five furniture items with at least 70 percent confidence. These outputs, along with the user's text input, are formatted into a structured prompt by GPT-4.1 nano. The prompt is then used to request an image from the DALL·E 3 API, which returns a generated image URL.

The generated image and analysis results are immediately displayed on the web interface. This includes the structured prompt, identified styles, and detected furniture, allowing users to receive clear feedback on how their input was interpreted.

To support this workflow, the system must integrate external AI models through modular

API connections. Each component—prompt generation, style extraction, object detection, and image rendering—should operate independently yet cohesively.

The UI must be simple and accessible, supporting users with little or no technical background. From input to output, the interface should offer a fast, smooth, and informative experience.

4.2.4 User-Related Requirement (사용자 관련 요구사항)

The HANCUT Web Service is intended for users without professional experience in interior design, such as small business owners aiming to visualize commercial spaces and general consumers interested in home decoration. These users are not expected to have prior knowledge of AI or design software.

To ensure accessibility, the interface must be intuitive and user-friendly. Users should be able to submit a short natural language description or a valid image URL with minimal instruction. The interface should clearly guide users through the input process with simple design elements and minimal interaction steps.

The system must provide fast and accurate results based on the given input. Prompt generation and image creation should operate in real time to maintain responsiveness and support user expectations for immediate visual output.

By combining easy input and timely feedback, the service enables non-expert users to engage with interior design tools in a simple and efficient way, expanding access to AI-based visualization.

4.3 Realistic Constrains (현실적 제한조건)

4.3.1 Constraints for Available Resources (가용 자원에 대한 제한조건)

The HANCUT Web Service project must consider cost, development infrastructure, and team capacity. Since image generation relies on the OpenAI API, each team member requires at least $5 in API credits, which should be planned as a recurring expense.

Development will use a Node.js frontend and Python backend within a virtual environment. The system must operate solely on CPU resources without GPU acceleration. Accordingly, AI models must be optimized for CPU use, and a minimum of 8GB RAM is required for stable performance.

All team members should have prior experience integrating vision-language models via API, particularly for text-to-image tasks involving SIGLIP and OpenAI services. This ensures the team can proceed without additional training.

The entire development process, including implementation and testing, must be completed within a single academic semester. Therefore, the scope and system design must align with this schedule and available resources.

4.3.2 Constraints for Minium Performance (최소 성능에 대한 제한조건)

To ensure a responsive and user-friendly experience, the HANCUT Web Service must meet the following minimum performance criteria:

- Prompt Generation Time

The system must complete all AI processing—including style extraction (SIGLIP), object detection (Faster R-CNN), and prompt structuring (GPT-4.1 nano)—within 10 seconds of receiving user text input and reference image URL.

- Image Generation Time

The composed domain-specific prompt must be processed by the DALL·E 3 API, and the generated image must be displayed in the web interface within 20 seconds.

- Module Latency

Each core module, including prompt creation, image generation, and UI rendering, must respond within 20 seconds to maintain interactive usability.

- Visual Fidelity of Furniture Elements

At least 50% of the furniture items detected from the reference image must be reflected in the generated image, ensuring perceptual accuracy.

These constraints are essential for sustaining real-time feedback, minimizing user frustration, and maintaining the credibility of AI-assisted design generation.

4.3.3 Constraints for Security (보안에 대한 제한조건)

The system implements strict access controls to prevent unauthorized interactions with core components and data flows. All user inputs are subject to validation to prevent the execution of malicious code. Invalid or malformed entries are rejected, and users are prompted to resubmit corrected inputs to maintain system integrity.

Request activity is temporarily logged for monitoring purposes but is not stored permanently. This non-retention approach supports user privacy by minimizing the risk of data exposure and ensuring compliance with modern data protection standards.

To maintain reliability, the system includes built-in error detection and recovery mechanisms. In the event of unexpected faults, critical functions remain operational and service continuity is

preserved.

4.3.4 Constraints for Compatibility (호환성에 대한 제한조건)

To ensure broad usability and long-term maintainability, the system must be compatible with widely used operating systems such as Windows and macOS. It should also operate reliably across major web browsers including Chrome and Firefox, providing a consistent user experience regardless of platform.

The architecture is modular, allowing independent development, testing, and updating of key system components like prompt generation, image analysis, and synthesis. This structure facilitates future upgrades and helps isolate issues without disrupting the entire service.

In addition, the system utilizes standardized RESTful APIs to support seamless integration with external AI services and third-party platforms. This enhances flexibility, enabling future migration or extension of functionalities without significant reengineering. These design choices ensure the system remains adaptable to evolving technical environments and user needs.

4.3.5 Constraints for Ethics (윤리성에 대한 제한조건)

The project strictly complies with intellectual property laws. All AI models, datasets, and libraries are open-source or publicly available, used under proper licenses. No proprietary or unauthorized data has been used in development, ensuring both legal and ethical use of resources.

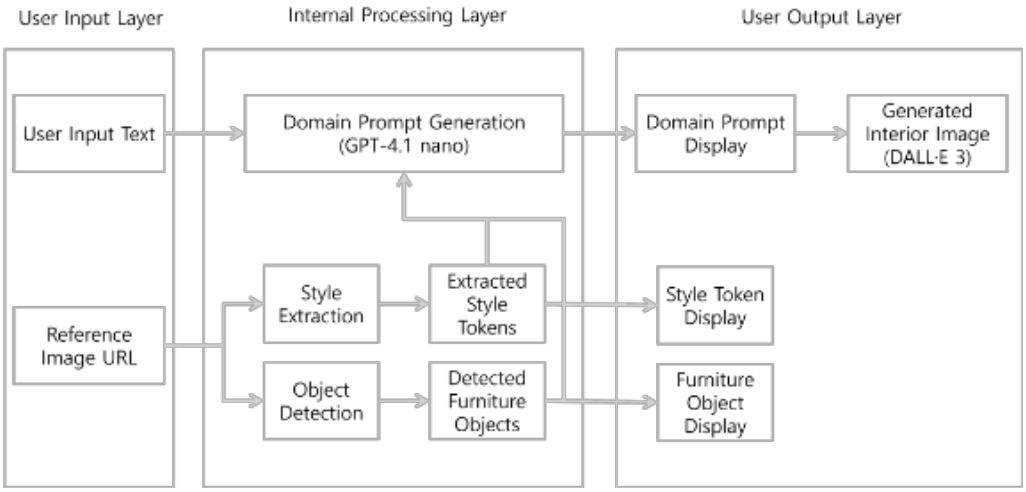
Only minimal user data is collected to fulfill system functions. Inputs are processed in real time without being stored or logged persistently, which protects user privacy and prevents misuse.

To maintain system safety, continuous testing and monitoring are in place. Design flaws or anomalies are resolved immediately through predefined procedures, reducing the risk of user harm.

5. Project Contents (과제 내용)

5.1 Design in Detail (상세 설계)

5.1.1 System Structure (시스템 구조)



[Figure 2] Overall Hancut Web Service Structure

The HANCUT Web Service is organized into three logical layers: User Input, Internal Processing, and User Output, each fulfilling a specific role in transforming user intent into a personalized interior image.

In the User Input Layer, users provide a natural language description of their interior preferences and a reference image via URL (in formats like JPEG, PNG, or WEBP). These inputs serve as the foundation for generating a customized prompt and image.

Within the Internal Processing Layer, the reference image is processed through two models. A fine-tuned SIGLIP model extracts the top three interior style tokens, while a pretrained Faster R-CNN identifies up to five furniture objects with a minimum 70% confidence. Alongside the user's text input, these elements are passed into the Domain Prompt Generation module using GPT-4.1 nano, which outputs a structured prompt designed for generative image synthesis.

The User Output Layer presents this prompt along with the extracted style and object information. The prompt is then sent to the DALL·E 3 API, which returns a generated interior image URL. This image is rendered immediately on the web platform, providing users with visual feedback.

The entire system is designed for modularity and real-time performance, completing prompt processing within 10 seconds and image rendering within 20 seconds. Its clear separation of input, processing, and output supports efficient debugging, scalability, and user-centric feedback.

5.1.2 Module Specification (모듈 설계)

The system consists of multiple operational modules, categorized into two domains: LLM for handling text-based inputs and VISION for processing reference images. The front-end collects user inputs and sends them to the back-end via RESTful APIs. The back-end processes the input and returns results, which are then visualized in the user interface.

1) LLM Domain Modules

- Domain Prompt Generation

-Module Names: generatePrompt (FrontEnd), generate_prompt (BackEnd)
-Input: Required user input text; optionally, style and object tokens
-Logic: The module receives input text and (optionally) style/object sequences, combines them with a predefined system prompt, and sends a POST request to an external LLM API (GPT-4.1 nano) to generate a domain-specific prompt.
-Output: A domain-optimized prompt string
-Input/Output Data Type: String → String
-API Endpoint: POST /api/llm/generate-prompt
-Dependencies: User Input Text (required); Style and Object tokens (optional)

- Generated Interior Image

-Module Names: generateImage (FrontEnd), generate_image (BackEnd)
-Input: Domain-specific prompt (required)
-Logic: Sends the prompt to the DALL·E 3 API to generate an image
-Output: URL of the generated image
-Input/Output Data Type: String → String
-API Endpoint: POST /api/llm/generate-image
-Dependencies: Dependent on Domain Prompt Generation

2) VISION Domain Modules

- Style Extraction

-Module Names: extractStyle (FrontEnd), extract_style (BackEnd)
-Input: Reference image URL
-Logic: Downloads the image, processes it via SIGLIP, and extracts interior style tokens
-Output: List of extracted style tokens
-Input/Output Data Type: String → List
-API Endpoint: POST /api/vision/extract-style
-Dependencies: Dependent on Reference Image URL

- Object Detection

-Module Names: detectObjects (FrontEnd), detect_objects (BackEnd)

-Input: Reference image URL
-Logic: Uses Faster R-CNN to detect furniture and decorative items
-Output: List of detected object tokens
-Input/Output Data Type: String → List
-API Endpoint: POST /api/vision/detect-objects
-Dependencies: Dependent on Reference Image URL

5.2 Implementation (구현)

5.2.1 Implementation Result (구현 결과)

(Writing Guideline) Present the detailed explanation of your system implementation results with main pictures or screen shots.



[Figure 3] Home Screen



[Figure 4] User Input Screen



[Figure 5] Analysis Screen



[Figure 6] Analysis Result Screen



[Figure 7] Prompt Result Screen



[Figure 8] Image Result Screen

Upon launching the HANCUT service at localhost:3000, the home screen appears as shown in [Figure 3]. The explanation below follows the ALL-IN-ONE Service workflow, which integrates all major functions.

In [Figure 4], users input a brief design description and a reference image URL. Once submitted, the system performs image analysis, as illustrated in [Figure 5], showing the uploaded image alongside the analysis process.

When the analysis is complete, [Figure 6] displays the extracted style keywords and detected furniture objects with confidence scores, together with the user's original text input.

Clicking the Generate Prompt button leads to [Figure 7], where a domain-specific prompt is shown. The final step, Generate Image, sends this prompt to the DALL·E 3 API and returns a generated interior image, which is presented in [Figure 8].

5.2.2 Implementation Tools (구현 도구)

This project integrated various frontend, backend, and AI tools to build an AI-powered interior design image generation platform. Each tool was selected for its performance, compatibility, and ease of use.

1) Frontend Stack

- TypeScript: Chosen for its static typing and reliability, TypeScript improved code safety and maintainability while handling text/image input and rendering results.

- Material-UI: Provided a wide range of responsive and customizable UI components, allowing for quick development of a clean and user-friendly interface.

2) Backend Stack

- FastAPI (Python 3.10.17): FastAPI supported high-performance asynchronous API handling and simplified unit testing with auto-generated Swagger documentation. Python 3.10.17 was used in a standalone environment to prevent module conflicts and ensure system compatibility.

- OpenAI API: GPT-4.1 nano generated structured prompts from user inputs with high semantic accuracy at a low cost. DALL·E 3 was used to generate high-quality interior images. Both APIs enabled scalable deployment without local model hosting.

- Hugging Face Transformers & PyTorch: Hugging Face was selected for its wide availability of pretrained models and smooth CI/CD integration. PyTorch served as the core framework for model training and inference.

3) AI Models Used

- SigLIP: A vision-language model used to extract interior style keywords from reference images. It was fine-tuned on 11,900 labeled interior images across 19 styles (Kaggle dataset) to improve classification accuracy.

- Faster R-CNN: Used to detect furniture objects in reference images and extract metadata to enrich prompt generation. This model was selected for its ability to run inference within 10 seconds in CPU-only environments, making it suitable for lightweight web deployment.

By combining these tools, we developed a modular, cost-efficient, and user-accessible platform for AI-driven interior image generation, requiring no GPU infrastructure and supporting seamless web-based interaction.

6. Evaluation

6.1 Metrics for evaluation

Evaluation Item	Unit	Reference Data	Evaluation Result					Evaluation Method
			Very Poor (1)	Poor (2)	Normal (3)	Good (4)	Very Good (5)	
Style Consistency Accuracy	%	50%	30	40	50	60	70	Extract style keywords from both images and calculate match ratio (30 times)
Object Consistency Accuracy	%	50%	45	50	55	60	65	Detect objects in both images and calculate match ratio (30 times)

6.2 Performance evaluation result

Evaluation Item	Measurement Result	Evaluation Result
Style Consistency Accuracy	44.44%	Poor
Object Consistency Accuracy	60.56%	Good

6.3 Verification of design constraints

1) Style Consistency Accuracy : Style accuracy was evaluated through 30 repeated tests. Although the average result was relatively low (44.44%), this does not fully reflect the system's actual performance.

According to expert mentoring feedback, style features are encoded as high-dimensional embedding vectors, and simple keyword-based comparisons may underestimate true alignment. Additionally, when users explicitly specify a style in their prompt, the system prioritizes that over the visual style extracted from the reference image. This behavior likely contributed to the lower score, despite the system functioning as intended.

2) Object Consistency Accuracy : This metric assesses how well the furniture and interior objects from the reference image are preserved in the generated image. The average score (60.56%) indicates a reasonable level of consistency, suggesting that the system reliably reflects key spatial elements. This result can be interpreted as an indicator of the system's practical performance and output trustworthiness.

6.4 User evaluation result

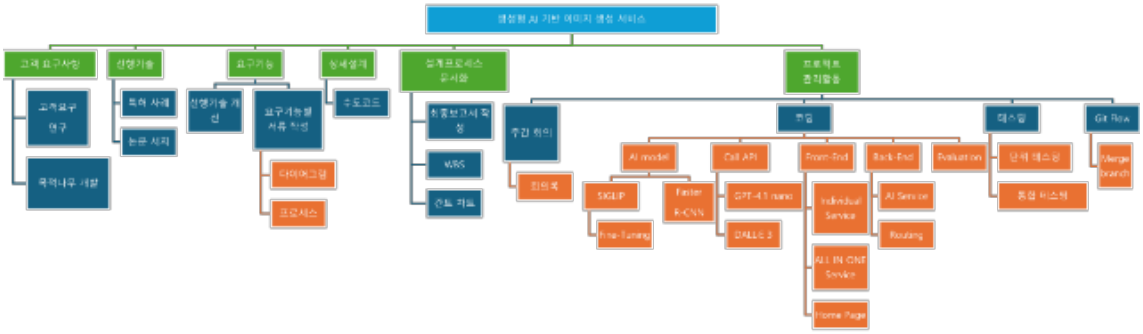
1) UI Intuitive : The interface is simple and intuitive, so even non-expert users can easily create interior design images. All key functions, including input and result viewing, are integrated into a single page for greater convenience.

2) Prompt Completion via LLM : When user input is vague or incomplete, the system utilizes a Large Language Model (LLM) to intelligently infer and fill missing components. This AI-assisted feature ensures that generated images still contain appropriate and coherent interior elements, improving output quality even in low-input scenarios.

3) Practical Usability : The streamlined workflow supports quick iterations and effective idea visualization, making the system especially useful for small business owners and other users without interior design expertise. The tool significantly lowers the barrier to accessing design guidance and reduces the need for repeated consultations.

7 Tasks and Task Assignments (과업 및 과업 분담)

7.1 Work Breakdown Structure (과업 분해)



[Figure 9] Work Breakdown Structure

7.2 Task Assignment and Project Timeline (과업 분담 및 과제 진행)



[Figure 10] Timeline

Tasks	Detail Description	Weight (1-10)	조양진 (%)	하유경 (%)	오진호 (%)	Total (%)
Project Lead	Make meetings	2	50	25	25	100
Meeting	Meeting Facilitation	3	50	25	25	100
Minutes	Meeting Minutes Preparation	4	10	10	80	100
Presentation	Final Presentation	7	100	0	0	100
Report	Document Preparation	6	40	20	40	100
Individual Sum			**1230**	**285**	**685**	

[figure 11] Work Division Contribution

8. Anticipated Results and Contributions (예상 결과 및 기대 효과)

Upon completion, the project successfully developed the HANCUT service—an integrated web platform that combines LLM, vision models, and diffusion-based generative AI to generate customized interior designs. The platform is released as open-source software, enabling public access and further development. To contribute to the AI and design community, the team has also shared a Drive link containing fine-tuned SIGLIP model weights in Tensor format, supporting research on interior style extraction.

- Impact on the Market

HANCUT enables small business owners to visualize and refine their interior designs without relying on professional consultants. This low-cost, AI-powered solution significantly reduces design planning time and cost, offering practical support to entrepreneurs with

limited budgets.

In 2024, approximately 1.18 million startups are expected to launch in South Korea, with 240,000 estimated to be small businesses (source: e-나라지표). Professional interior design services typically cost between KRW 220,000 and 400,000 (source: 숨고), posing a financial challenge for small businesses.

In contrast, HANCUT provides an affordable alternative. If just 1% (2,400 businesses) subscribe at KRW 10,000 per month, the service could generate KRW 280 million in annual revenue, demonstrating strong commercial potential.

Beyond commercial use, HANCUT is also applicable to residential interior design. As Korea's interior market expanded from KRW 7 trillion in 2008 to KRW 18 trillion in 2023, the "home-decorating" trend continues to grow. By lowering the barrier to creating design blueprints, the service aligns with consumer trends and encourages the purchase of interior products, further stimulating related markets.

The platform also helps lower the entry barrier for entrepreneurship and business re-entry, potentially contributing to local commercial revitalization, job creation, and broader positive social impact.

- Impact on Technology

By open-sourcing its interior prompt generation technology, the project establishes a global reference point for generative AI applications in interior design. This not only promotes innovation in Korea but also supports broader adoption in international markets.

- Impact on Human Life

HANCUT offers an accessible design tool for everyone, regardless of their expertise or background. This democratization of interior design promotes personal creativity, strengthens one's sense of space ownership, and contributes to a more fulfilling and enriched lifestyle.

9. References (참고 서적)

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10. Budget (예산)

- OpenAI API Resource

To support prompt structuring and image generation, the project utilized OpenAI's API services. Each team member contributed a minimum credit of $5 to ensure adequate access and improve the overall quality and completeness of the project.

1) GPT-4.1-nano API Usage

The GPT-4.1 nano API was used to generate domain-specific prompts that harmonize user-provided text with visual elements such as extracted interior styles and detected objects. During development, the model was executed repeatedly to validate results. This process consumed over 2 million input tokens and more than 4 million output tokens, leading to a minimum estimated cost of approximately $1.80 (calculated as $0.20 for input and $1.60 for output tokens).

Text tokens — Price per 1M tokens · Batch API price

Model	Input	Cached input	Output
gpt-4.1 ↳ gpt-4.1-2025-04-14	$2.00	$0.50	$8.00
gpt-4.1-mini ↳ gpt-4.1-mini-2025-04-14	$0.40	$0.10	$1.60
gpt-4.1-nano ↳ gpt-4.1-nano-2025-04-14	$0.10	$0.025	$0.40

[Figure 11] GPT-4.1-nano API Input/Output Token Costs

2) DALL·E 3 API Usage

The DALL·E 3 API was used to generate high-resolution interior images based on the structured prompts. Each image was rendered at 1792×1024 resolution to meet visual quality expectations. To iteratively validate prompt quality, the API was called over 50 times throughout the project. At a rate of $0.08 per image, this resulted in a minimum cost of $4.00.

Model	Quality	1024x1024	1024x1536	1536x1024
GPT Image 1	Low	$0.011	$0.016	$0.016
	Medium	$0.042	$0.063	$0.063
	High	$0.167	$0.25	$0.25
Model	**Quality**	**1024x1024**	**1024x1792**	**1792x1024**
DALL·E 3	Standard	$0.04	$0.08	$0.08
	HD	$0.08	$0.12	$0.12

[Figure 12] DALL·E 3 API Costs